Meta-analysis results

Evolution of ISUP Grade Group 1 Detection Across Prostate Cancer Screening Strategies

Analysis performed per `protocol.md` §8 (Statistical Analysis) and §9 (Data Synthesis). R 4.6.0, `meta` 8.5.0, `metafor` 5.0.1. Script: meta_analysis.R. Full numeric output: output/results.txt.

1. Included data

Nine randomized trials, 18 arms, 782,328 men randomized/invited. All nine parse internally consistently: ISUP GG1 + ISUP ≥2 ≤ any PCa in every arm, and the reported "GG1 / any PCa" percentages reproduce exactly from the component counts.

The trials divide into two contrast families, which are **never pooled into a single interpreted estimate** (protocol §8/§9; also required by the PROSA extraction note):

Family	Trials	Contrast
A	ERSPC 2009, PLCO 2009, CAP 2018, PROBASE 2022, ProScreen 2024	Screening invitation vs no screening / usual care
B	STHLM3-MRI 2021, MVP 2022, GÖTEBORG-2 2024, PROSA 2026	Diagnostic / biopsy strategy, both arms screened

Arms were classified by **the triage test that triggers work-up**:

Strategy	Arms (k)
No screening / usual care	ERSPC-C, PLCO-C, CAP-C, ProScreen-C (4)
PSA + DRE	PLCO-I (1)
PSA	ERSPC-I, CAP-I, PROBASE-I, PROBASE-C, STHLM3-C, MVP-C (6)
PSA → MRI	GÖTEBORG-2-I, GÖTEBORG-2-C, PROSA-C (3)
Biomarker → MRI	ProScreen-I, STHLM3-I (2)
MRI-first	MVP-I, PROSA-I (2)

2. Primary outcome — ISUP GG1 detection

2.1 Comparative (risk ratios, ITT denominators) — Figure 1

Random effects, inverse variance, DerSimonian–Laird τ, Hartung–Knapp CI, 0.5 correction for zero cells.

Family	k	RR (95% CI)
A. Invitation vs no screening	5	**1.76 (1.09 – 2.84)**
B. Diagnostic / biopsy strategy	4	**0.56 (0.15 – 2.08)**
Test for subgroup differences		Q = 6.51, df = 1, **p = 0.011**

Inviting men to screening increases GG1 detection by roughly three quarters (all five family-A trials point the same way: RR 1.27 to 7.49). Within already-screened populations, modern pathways point toward *reduced* GG1 detection, but the estimate is imprecise and directionally inconsistent (STHLM3-MRI 0.31 and GÖTEBORG-2 0.43 vs PROSA 2.32 and MVP 1.03) — the two large trials drive the reduction, the two small ones oppose it. The families differ significantly.

Heterogeneity is extreme throughout: overall Q = 388.3, df = 8, p < 0.0001, I^2 = 97.9%. Leave-one-out: family-level conclusions hold; no single trial drives the family-A result (omitting ERSPC gives the largest shift, overall RR 1.17 → 1.02).

2.2 Grade mix — GG1 as a share of all detected cancers — Figure 4

This is the more comparable outcome, because it is internally normalized within each trial and so is not distorted by differences in follow-up duration, screening rounds, or age.

Family	k	RR (95% CI)	Excluding PROBASE
A. Invitation vs no screening	5	1.08 (0.64 – 1.82)	**1.21 (1.00 – 1.47)**
B. Diagnostic / biopsy strategy	4	**0.57 (0.30 – 1.06)**	0.57 (0.30 – 1.06)
Subgroup difference		p = 0.019	p = 0.00025

Screening invitation shifts the detected-cancer mix *toward* low grade (ERSPC 1.32, CAP 1.30, PLCO 1.09 — all significant). Modern diagnostic pathways shift it *away* from low grade by about 43% relative, just short of conventional significance.

2.3 Pooled proportions by strategy (GLMM) — Figures 6 and 8

GG1 as % of all detected prostate cancers — the evolution signal:

Strategy	k	GG1 share of detected PCa (95% CI)
PSA + DRE	1	65.7% (64.1 – 67.3)
PSA	6	49.4% (35.1 – 63.9)

Strategy	k	GG1 share of detected PCa (95% CI)
PSA → MRI	3	44.3% (33.9 – 55.2)
No screening / usual care	4	39.8% (23.8 – 58.4)
MRI-first	2	31.7% (19.4 – 47.3)
Biomarker → MRI	2	16.4% (13.2 – 20.1)
Between-strategy difference		$\chi^2 = 325.0$, df = 5, $p < 0.0001$

A clear gradient: roughly two thirds of screen-detected cancers were GG1 in the PSA + DRE era, about half under PSA alone, and one sixth to one third under biomarker- or MRI-triaged pathways. This is the quantitative backbone of the review's thesis. Heterogeneity within the two MRI/biomarker groups is nil ($I^2 = 0\%$, Q p = 0.43 and 0.60), so those two estimates are the most trustworthy in the table.

GG1 detection rate per man screened (denominators = men actually screened where reported):

Strategy	k	GG1 per 100 screened (95% CI)
PSA + DRE	1	6.96% (6.69 – 7.24)
MRI-first	2	2.11% (1.23 – 3.61)
PSA → MRI	3	1.46% (0.87 – 2.45)
No screening / usual care	4	1.20% (0.33 – 4.30)
PSA	6	0.60% (0.08 – 4.14)
Biomarker → MRI	2	0.48% (0.38 – 0.60)

> **These per-screened rates should not be read as strategy effects.** The PSA group has $\tau^2 = 5.90$ and spans a 120-fold range (ERSPC-I 7.22% to PROBASE-I 0.06%), so its pooled estimate of 0.60% lies *below almost every arm contributing to it* — a mathematical artefact of logit-scale pooling under extreme heterogeneity, not a finding. The dominant drivers are follow-up duration (9–10 years in ERSPC/PLCO/CAP vs a single round elsewhere), number of screening rounds, and age at screening (45 in PROBASE), not the screening strategy. Only the two homogeneous groups (Biomarker → MRI, $I^2 = 0\%$; MRI-first, $I^2 = 0\%$) support direct interpretation.

3. Secondary outcomes

3.1 ISUP ≥2 (csPCa) detection — Figures 2 and 7

Family	k	RR (95% CI)
A. Invitation vs no screening	5	1.16 (0.76 – 1.77)
B. Diagnostic / biopsy strategy	4	1.29 (0.57 – 2.93)
Subgroup difference		p = 0.71

No detectable difference in clinically significant cancer detection in either family, and no difference between families. Read alongside §2.1, this is the central asymmetry: screening invitation raised GG1 detection by ~76% while csPCa detection was essentially unchanged (RR 1.05, 0.99, 1.04 in ERSPC, PLCO, CAP respectively) — i.e. the excess detection was almost entirely low grade.

Pooled csPCa per 100 men screened: MRI-first 4.55% (3.16 – 6.52), PSA + DRE 3.39%, Biomarker → MRI 2.37% (2.14 – 2.62), PSA → MRI 1.94% (1.72 – 2.19), no screening 1.69%, PSA 0.47%. The same denominator caveat as §2.3 applies to the PSA and no-screening rows.

3.2 Biopsies performed — Figures 5 and 9

Comparative analysis is limited to the 5 trials reporting biopsies in **both** arms (ERSPC, CAP and ProScreen report the intervention arm only; PLCO reports neither):

Family	k	RR (95% CI)
A (PROBASE only)	1	3.24 (2.24 – 4.68)
B. Diagnostic / biopsy strategy	4	0.71 (0.24 – 2.15)
Subgroup difference		p = 0.00013

Within family B the direction splits sharply by design: the two triage-tightening trials cut biopsies by more than half (STHLM3-MRI 0.48, GÖTEBORG-2 0.43) while the two MRI-first trials increased them (PROSA 2.06, MVP 0.88). Pooled biopsies per 100 men screened: MRI-first 10.41% (8.23 – 13.08), PSA → MRI 5.70%, PSA 3.80%, Biomarker → MRI 3.72% (3.27 – 4.24).

3.3 Curative treatment — not analysable

The protocol lists "percentage of patients receiving curative treatment over follow-up" as a secondary outcome. These data are not present in the extraction and this outcome could not be analysed. Adverse-event / procedural-harm data (protocol §5, §9) are likewise absent from the extraction.

4. Heterogeneity, sensitivity, publication bias

Heterogeneity was significant by Cochran's Q ($p < 0.05$) in every pooled analysis, with I^2 from 68.9% to 99.9%. Per protocol §8 this triggered exploratory analysis, which is what the family and strategy stratifications above provide.

Leave-one-out was run on every pooled estimate (see `output/results.txt`). No conclusion depends on a single trial.

Targeted sensitivity analyses:

Analysis	Effect
Exclude **PROBASE** (delayed-screening comparator; control csPCa = 0 was *inferred by the extractor, not reported*)	Family A GG1 RR 1.76 → 1.67 (1.12 – 2.49); family A grade-mix RR 1.08 → 1.21 (1.00 – 1.47), subgroup difference strengthens to p = 0.00025
Exclude **ERSPC** (Gleason counts back-calculated from published percentages)	Family A GG1 RR 1.76 → 1.50 (0.96 – 2.34) — loses significance
MVP / PROSA on randomized rather than analysed denominators	No material change (GG1 RR 1.167 both ways)
Continuity correction **0.1 / 0.5 / 1.0** on the single zero cell	Family A csPCa RR 1.14 / 1.16 / 1.17 — robust
Proportions on **ITT rather than per-screened** denominators (Figs. S1, S2)	Reorders the per-man rate tables; the grade-mix ranking in §2.3 is unaffected, as it uses detected cancers as denominator

Publication bias was not assessed. The protocol restricts Peters' regression and funnel plots to analyses with ≥10 studies; the largest here has k = 9. Small-study effects cannot be excluded — relevant given that the two trials opposing the family-B GG1 reduction (MVP, PROSA) are also the two smallest.

Protocol deviation: §8 specifies a 0.5 continuity correction for zero cells in both the risk-ratio and proportion analyses. It was applied to the risk-ratio models only. The GLMM proportion models use a logit-normal likelihood that accommodates zero counts directly, and adding 0.5 would bias those estimates; this is standard practice.

5. Interpretation

1. **Invitation to screening increases low-grade detection without a matching increase in significant cancer** (GG1 RR 1.76 vs csPCa RR 1.16), and shifts the detected-cancer mix toward GG1 (RR 1.21 excluding PROBASE). This is overdiagnosis quantified.
2. **Modern triage reverses the grade mix.** GG1 fell from 65.7% of detected cancers under PSA + DRE and 49.4% under PSA to 31.7% (MRI-first) and 16.4% (biomarker → MRI), $p < 0.0001$ between strategies; within-trial RR 0.57 in family B. The two biomarker → MRI arms agree closely ($I^2 = 0\%$).
3. **The cost side differs by strategy type.** Biomarker-triaged pathways reduce GG1 detection *and* biopsies (STHLM3-MRI, GÖTEBORG-2: biopsy RR 0.48 and 0.43). MRI-first pathways achieve the lowest GG1 share but at the highest biopsy rate (10.4 per 100 screened) — though on only 615 men across two small, early-terminated or single-round trials.
4. **The main threat to these conclusions is between-trial comparability, not sampling error.** Trials differ by 9–10 years vs one round of follow-up, by age 45 vs 55–69, and in control-arm contamination (PLCO's control arm had 40–52% PSA uptake, which biases its RRs toward 1). Per-man detection rates are therefore not interpretable across trials; the grade-mix outcome is, and it carries the review's message.

6. Output files

File	Contents
output/results.txt	Complete numeric output, all models, all leave-one-out runs
output/table1_descriptives.csv	Arm-level extracted data with derived rates
output/table2_pooled_estimates.csv	All pooled estimates with Q, df, p, I^2
`output/fig1` – `fig5` (pdf + png)	Risk-ratio forest plots: GG1, csPCa, any PCa, grade mix, biopsies
`output/fig6` – `fig9` (pdf + png)	Proportion forest plots by strategy: GG1/screened, csPCa/screened, GG1 share, biopsies/screened
`output/figS1` – `figS2` (pdf + png)	Sensitivity: proportions on ITT denominators
arm_level_data.csv	Analysis dataset, with per-arm provenance and caveat flags